

Question number	Answer	Additional guidance	Mark
(a)	An answer that includes the following point: - A is the correct answer - B is not the correct answer as adenosine diphosphate is a high energy phosphate compound - C is not the correct answer as adenosine triphosphate is a high energy phosphate compound - D is not the correct answer as cholesterol is a component of membranes		(1)

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(b)	An explanation that includes two of the following points: - stimulus initiates change in (voltage / membrane potential)(1) - sodium ion channels open / sodium ions rush in causing membrane potential to become less negative (1) - if above threshold potential depolarisation initiated (1)	Stimulus cause change from resting potential	(2)

Question number	Answer	Additional guidance	Mark
(c)(i)	A calculation showing the following steps: manipulation of equation to get m as subject (1) calculation of gradient for myelinated fibres using equation (1) calculation of percentage difference between gradients (1)	$m = (y - c) \div x$ $(120 - 0) / 20 = 6$ $(6 - 0.5) \div 0.5 \times 100 = 1100\%$ accept range 1050-1150 Correct answer gains full marks	(3)

Question number	Answer	Additional guidance	Mark
(c)(ii)	An explanation that includes three of the following points: • myelin (covers axon) and acts as an insulator (1) • {myelinated fibres have regular gaps / nodes of Ranvier} along axon where action potential occurs (1) • nerve impulse jumps {from node to node/ in saltatory conduction} (1)	accept opening of gated channels at nodes of Ranvier where action potential occurs	(3)

Question number	Answer	Additional guidance	Mark
(d)	A description that includes four of the following points: • radioactive tracer added with glucose (and injected into blood stream) (1) • (carried to brain and) taken up by active (respiring) cells (1) • (sensors in)PET scanner detect radioactivity (in different regions of the brain) (1) • build up (2D or 3D) images to show where cells stimulated (by cocaine) are found (1) • the PET shows more regions taking up radio activity in the(cocaine) brain (1)	accept added to water accept gamma rays	(4)